Wasmir Chowdhury

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EDUCATION

Carnegie Mellon University — Pittsburgh, PA
Bachelor of Science in Electrical and Computer Engineering, GPA: 3.28 — May 2027

Relevant Coursework
Embedded Systems, Real-Time Operating Systems, Structure and Design of Digital Systems, Computer Systems

WORK EXPERIENCE

Course Teaching Assistant — August 2025 – Present
18-213: Introduction to Computer Systems (Carnegie Mellon University) — Pittsburgh, PA
- Mentor 10 students per semester in C programming, memory management, and concurrency through weekly office hours and group debugging sessions

Undergraduate Research Assistant — March 2024 – February 2025
CREATE Lab (Carnegie Mellon University) — Pittsburgh, PA
- Developed Python and bash scripts to automate data collection, testing, and validation of AI model outputs across large datasets
- Designed and prototyped a Raspberry Pi-based sensor system for real-world field deployment and data logging

PROJECT EXPERIENCE

Real-Time Operating System (RTOS) Kernel | C, ARM Assembly
- Built a preemptive RMS scheduler on ARM Cortex-M4 with UB-test-based admission control
- Implemented HLP/IPCP mutex locking in C to prevent priority inversion and deadlocks across concurrent tasks
- Wrote a PendSV context switcher in ARM Assembly to manage thread state and CPU register save and restore

STM32 Bare-Metal Peripheral Drivers | C, STM32, ARM Cortex-M
- Implemented interrupt-driven I2C, SPI, and UART device drivers from scratch by directly configuring peripheral registers on a resource-constrained microcontroller
- Debugged communication timing using GDB over JTAG and an oscilloscope to validate driver correctness

STM32 Bare-Metal Bootloader | C, ARM Assembly, STM32
- Wrote a bare-metal bootloader in ARM Assembly, building the interrupt vector table and reset handler from scratch
- Initialized the processor memory map by copying data from flash to SRAM and zeroing BSS using linker script symbols

RetroTron (Hardware Hackathon, Build18) | ESP32, FPGA, SystemVerilog
- Designed and implemented digital logic on an FPGA in SystemVerilog, integrating with an ESP32 microcontroller over a wireless link
- Brought up the full embedded system from scratch, managing peripheral setup and inter-device communication

QNX Real-Time Vibration Monitor (Hackathon) | Python, QNX RTOS, Raspberry Pi 5
- Built a real-time vibration monitoring system on Raspberry Pi 5 running QNX Neutrino RTOS, sampling an MPU6050 via I2C at 100Hz and classifying anomalies at the edge
- Implemented a pure Python DFT/FFT pipeline for spectral analysis, extracting RMS amplitude and dominant frequency shift to detect mechanical faults without external dependencies

LEADERSHIP EXPERIENCE

EMERGE — July 2024 – Present
Mentor, Carnegie Mellon University — Pittsburgh, PA
- Facilitate monthly group meetings with mentees to offer academic, personal, and professional guidance

TECHNICAL SKILLS

Embedded Systems: STM32, ARM Cortex-M, RTOS (RMS, HLP/IPCP), Bare-Metal Programming, I2C, UART, SPI, Interrupts, GDB, JTAG
Software: C, C++, ARM Assembly, Python, SystemVerilog, Git/GitHub, Vivado, Fusion360 PCB